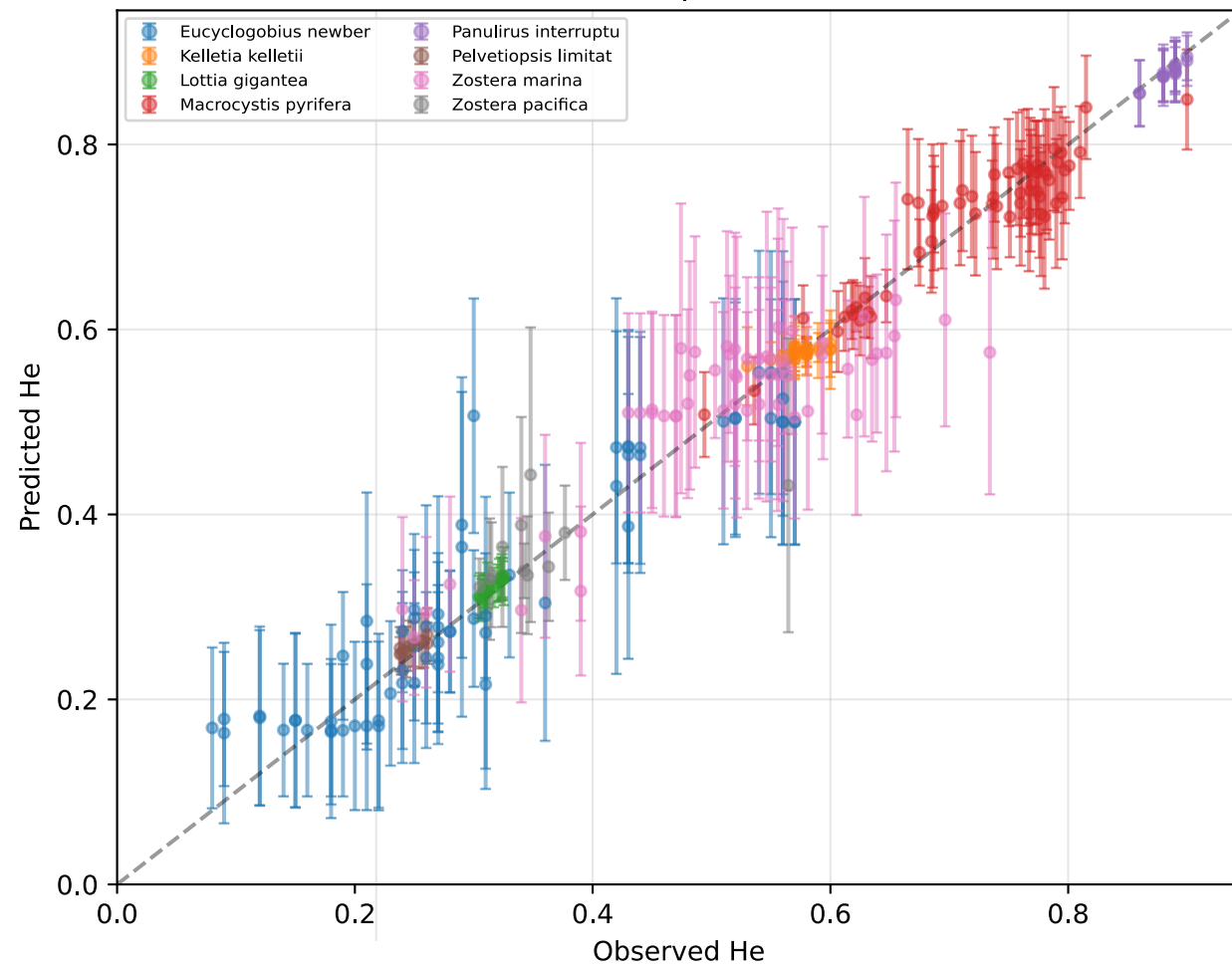


Leave-One-Out Predictions (CNP) — overall fit vs within-species skill

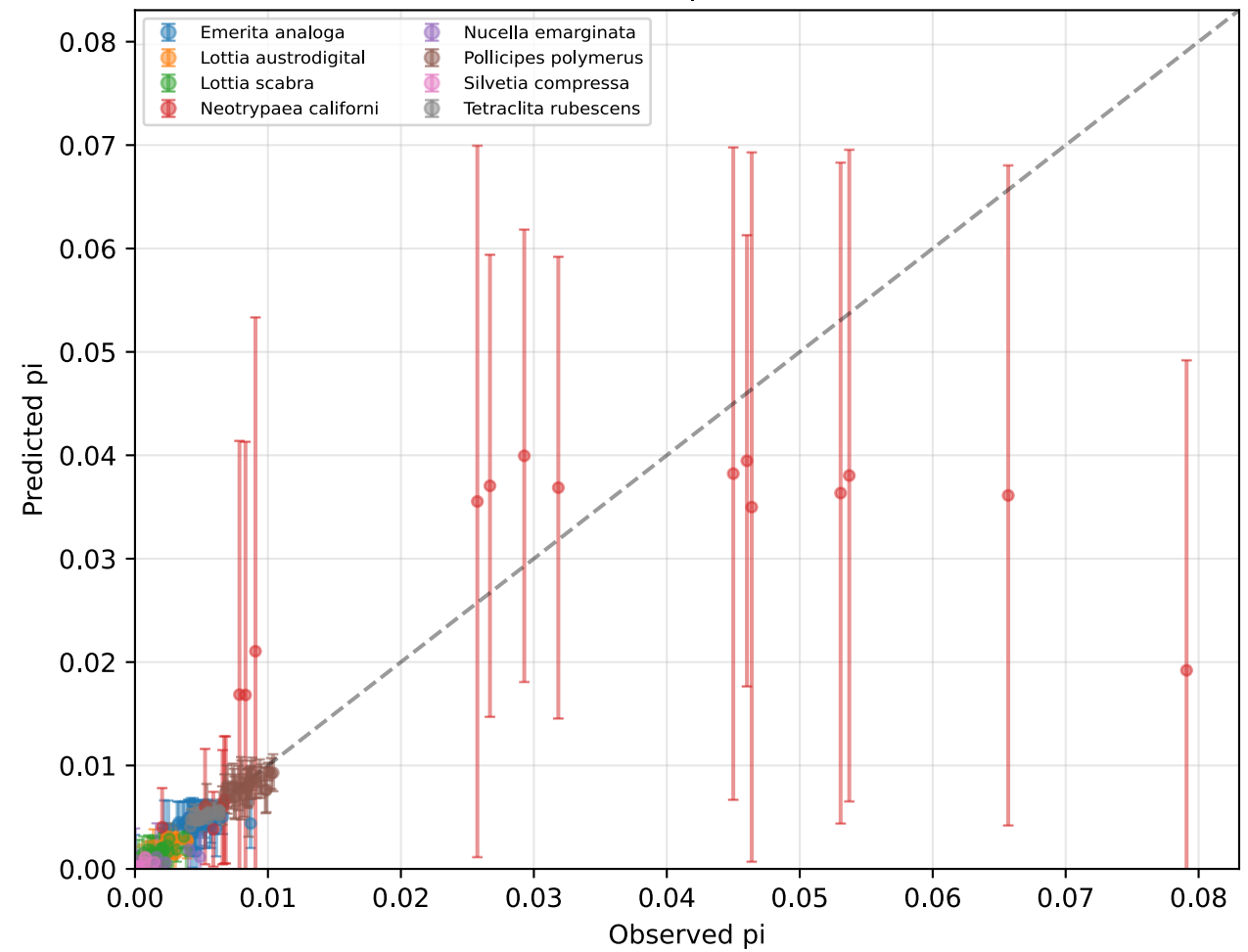
(a) Heterozygosity species

$R^2 = +0.96$ (mostly between-species) | predicted spread = 96% of observed



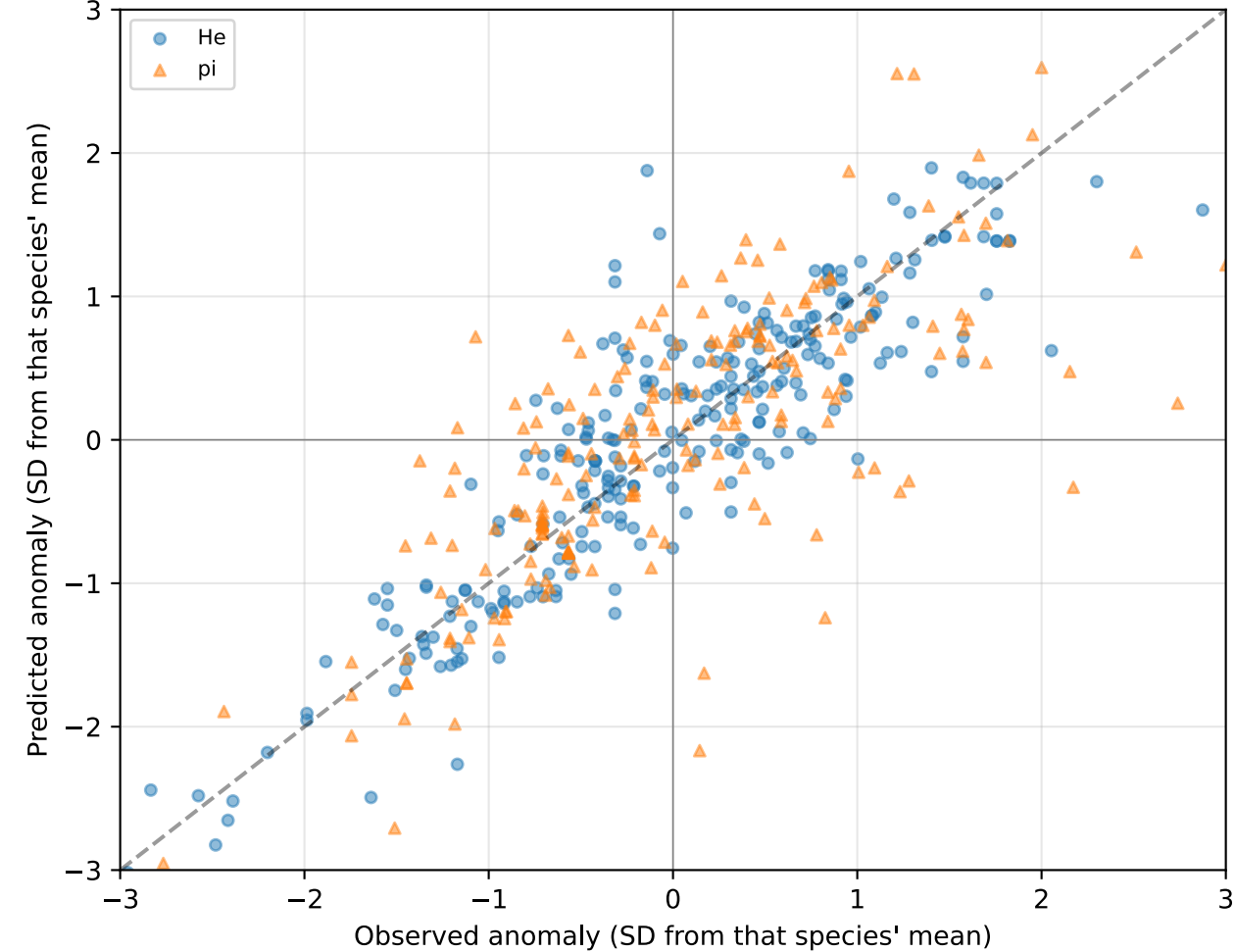
(b) Nucleotide diversity species

$R^2 = +0.74$ (mostly between-species) | predicted spread = 75% of observed



(c) Within-species skill

pooled $r = +0.83$ | median per-species $r = +0.80$ (0/16 negative)



(d) Global CNP pre-training Within species skill

$n=6245$ populations across 1108 species | $R^2=0.81$, RMSE=0.098

